Linear Stability Analysis of Continuum Equation

Jinghao Chen
May 3, 2020

Recall the continuum limit of discrete model is the following:

$$\frac{\partial \rho}{\partial t} = \frac{\partial^2 K(\rho)}{\partial x^2}$$

where $K(\rho) = \rho(1 - 2\alpha\rho + \alpha\rho^2)$.

Now, consider

$$\rho(x,t) = \bar{\rho} + \varepsilon\tilde{\rho}(x,t)$$

with

$$\tilde{\rho}(x,t) = \sum_{k=-\infty}^{+\infty} \hat{\rho}_k(t) e^{ikx}$$

$$LHS = \frac{\partial \rho}{\partial t} = \varepsilon \sum_{k=-\infty}^{+\infty} \frac{\partial \hat{\rho}_k(t)}{\partial t} e^{ikx}$$

while for right hand side, we use Taylor expansion on K at $\bar{\rho}$, where

$$K(\bar{\rho} + \varepsilon\tilde{\rho}) = K(\bar{\rho}) + \varepsilon\tilde{\rho} \cdot K'(\bar{\rho}) + O(\varepsilon^2)$$

and then take it in:

$$\begin{aligned} RHS &= \frac{\partial^2 K(\rho)}{\partial x^2} \\ &= \frac{\partial^2}{\partial x^2}\big(K(\bar{\rho} + \varepsilon\tilde{\rho})\big) \\ &= \frac{\partial^2}{\partial x^2}\big(K(\bar{\rho}) + \varepsilon\tilde{\rho} \cdot K'(\bar{\rho}) + O(\varepsilon^2)\big) \\ &= \frac{\partial^2 K(\bar{\rho})}{\partial x^2} + \varepsilon K'(\bar{\rho}) \cdot \frac{\partial^2 \tilde{\rho}}{\partial x^2} + O(\varepsilon^2) \\ &= \varepsilon K'(\bar{\rho}) \cdot \frac{\partial^2 \tilde{\rho}}{\partial x^2} + O(\varepsilon^2) \end{aligned}$$

As $\tilde{\rho}(x,t) = \sum_{k=-\infty}^{+\infty} \hat{\rho}_k(t) e^{ikx}$, so $\frac{\partial^2 \tilde{\rho}}{\partial x^2} = -\sum_{k=-\infty}^{+\infty} \hat{\rho}_k(t) k^2 e^{ikx}$, hence

$$RHS = -\varepsilon K'(\bar{\rho}) \cdot \sum_{k=-\infty}^{+\infty} \hat{\rho}_k(t) k^2 e^{ikx} + O(\varepsilon^2)$$

Let's manage the equation by taking $LHS = RHS$, we get

$$\sum_{k=-\infty}^{+\infty} \frac{\partial \hat{\rho}_k(t)}{\partial t} e^{ikx} = -K'(\bar{\rho}) \cdot \sum_{k=-\infty}^{+\infty} \hat{\rho}_k(t) k^2 e^{ikx} + O(\varepsilon)$$

so

$$\sum_{k=-\infty}^{+\infty} e^{ikx} \left(\frac{\partial \hat{\rho}_k(t)}{\partial t} + k^2 K'(\bar{\rho}) \hat{\rho}_k(t) \right) = O(\varepsilon)$$

As the $O(\varepsilon)$ term should be neglectable during the linear analysis, the equation above is equivalent to the following:

$$\frac{\partial \hat{\rho}_k(t)}{\partial t} + k^2 K'(\bar{\rho}) \hat{\rho}_k(t) = 0$$

which becomes an ODE of $\hat{\rho}_k$ w.r.t t, and can be easily solved as

$$\hat{\rho}_k(t) = \hat{\rho}_k(0) \cdot e^{-k^2 K'(\bar{\rho}) t}$$

In order to make it stable, i.e. prevent the solution above from blow up as $t \to +\infty$, this requires

$$K'(\bar{\rho}) \geq 0$$

recall that

$$K'(\rho) = D(\rho) = 3\alpha \left(\rho - \frac{2}{3} \right)^2 + 1 - \frac{4}{3}\alpha$$

which is the diffusivity. That means our stable requirement agrees with positivity of the diffusion term, which implies our continuum limit equation $\frac{\partial \rho}{\partial t} = \frac{\partial^2 K(\rho)}{\partial x^2}$ is a stable nonlinear diffusion equation for $0 \leq \alpha < \frac{3}{4}$. Otherwise, say, for $\frac{3}{4} \leq \alpha \leq 1$, we can always find out an interval I_α in which the diffusion term is negative making our continuum equation be ill-posed. Specifically, that interval is

$$I_\alpha = \left(\frac{2\alpha - \sqrt{\alpha(4\alpha - 3)}}{3\alpha}, \frac{2\alpha + \sqrt{\alpha(4\alpha - 3)}}{3\alpha} \right)$$